Covalent Bonding and Lewis Structures

Learning goals:

Writing valid Lewis structures for molecular substances

Predicting molecular geometry from Lewis structures (VSEPR theory)

Understanding electronegativity and how this concept allows the distinction between polar bonds and non-polar bonds

Using Lewis structures to determine whether a molecule has a dipole moment or not

Using the octet rule to compute formal charges on atoms and multiple bonding between atoms

Covalent Bonding and Lewis Structures

(1) Lewis "dot" (electron) structures of valence electrons for atoms

(2) Use of Periodic Table to determine the number of "dots"

(3) Use of Lewis structures to describe the electronic structures of atoms and molecules

(4) Works best for covalent bonds and for elements in the first full row of the Periodic Table: H, He, Li, Be, B, C, N, O, F, Ne

(5) Works with restrictions for second full row of the Periodic Table and beyond: Na, Mg, Al, Si, P, S, Cl, Ar

Some issues about Lewis Structures to be discussed:

(1) Drawing “valid” Lewis structures which follow the “octet” rule (holds almost without exception for first full row)

(2) Drawing structures with single, double and triple bonds

(3) Dealing with isomers (same composition, different constitution)

(4) Dealing with resonance structures (same constitution, different bonding between atoms)

(5) Dealing with “formal” charges on atoms in Lewis structures

(6) Dealing with violations of the octet rule:

Molecules which possess an odd number of electrons

Molecules which are electron deficient

Molecules which are capable of making more than four covalent bonds

Lewis “dot-line” representations of atoms and molecules

(1) Electrons of an atom are of two types: ***core*** electrons and ***valence*** electrons. Only the valence electrons are shown in Lewis dot-line structures.

(2) The number of valence electrons is equal to the group number of the element for the representative elements.

(3) For atoms the first four dots are displayed around the four “sides” of the symbol for the atom.

(4) If there are more than four electrons, the dots are paired with those already present until an octet is achieved.

(5) Ionic compounds are produced by complete transfer of an electron from one atom to another.

(6) Covalent compounds are produced by sharing of one or more pairs of electrons by two atoms.

The ***valence capacity*** of an atom is the atom's ability to form bonds with other atoms. The more bonds the higher the valence.

The valence of an atom is not fixed, but some atoms have typical valences which are most common:

Carbon: valence of 4
Nitrogen: valence of 3 (neutral molecules) or 4 (cations)
Oxygen: valence of 2 (neutral molecules) or 3 (cations)
Fluorine: valence of 1(neutral molecules) or 2 (cations)

Covalent bonding and Lewis structures

(1) Covalent bonds are formed from sharing of electrons by two atoms.

(2) Molecules possess only covalent bonds.

(2) The bedrock rule for writing Lewis structures for the first full row of the periodic table is the octet rule for C, N, O and F: ***C, N, O and F atoms are always surrounded by eight valence electrons.***

(4) For hydrogen atoms, the doublet rule is applied: ***H atoms are surrounded by two valence electrons.***

3.4
Covalent Bonds and Lewis Structures

The Lewis Model of Chemical Bonding

- In 1916 G. N. Lewis proposed that atoms combine in order to achieve a more stable electron configuration.
- Maximum stability results when an atom is isoelectronic with a noble gas.
- An electron pair that is shared between two atoms constitutes a covalent bond.

Covalent Bonding in H_2

Two hydrogen atoms, each with 1 electron,

$$H\cdot \qquad \cdot H$$

can share those electrons in a covalent bond.

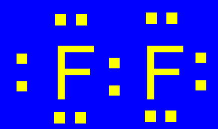
$$H:H$$

- Sharing the electron pair gives each hydrogen an electron configuration analogous to helium.

Covalent Bonding in F_2

Two fluorine atoms, each with 7 valence electrons,

can share those electrons in a covalent bond.



- Sharing the electron pair gives each fluorine an electron configuration analogous to neon.

The Octet Rule

In forming compounds, atoms gain, lose, or share electrons to give a stable electron configuration characterized by 8 valence electrons.

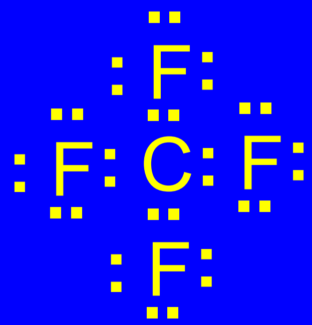
$$:\underset{\cdot\cdot}{\overset{\cdot\cdot}{F}}:\underset{\cdot\cdot}{\overset{\cdot\cdot}{F}}:$$

- The octet rule is the most useful in cases involving covalent bonds to C, N, O, and F.

Example

Combine carbon (4 valence electrons) and four fluorines (7 valence electrons each)

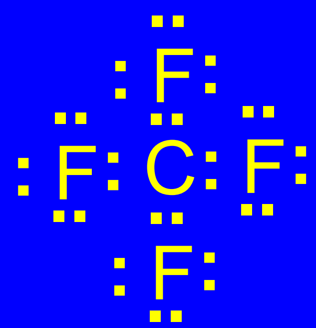
to write a Lewis structure for CF_4.

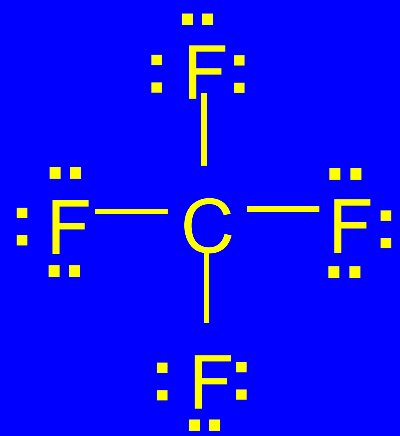


The octet rule is satisfied for carbon and each fluorine.

Example

It is common practice to represent a covalent bond by a line. We can rewrite



as 

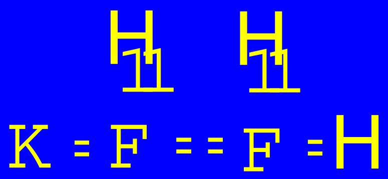
3.4 Double Bonds and Triple Bonds

Inorganic examples

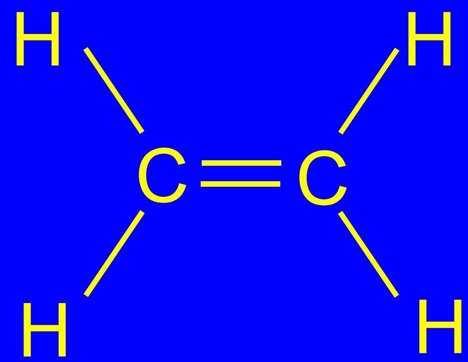
Carbon dioxide

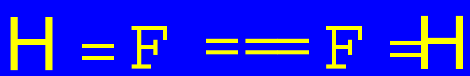
Hydrogen cyanide

Organic examples

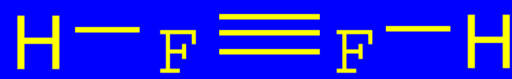


Ethylene





Acetylene

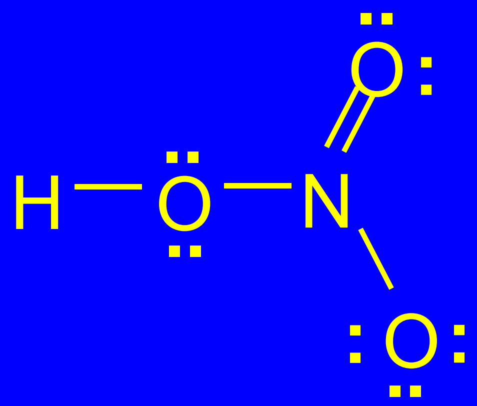


3.4
Formal Charges

- Formal charge is the charge calculated for an atom in a Lewis structure on the basis of an equal sharing of bonded electron pairs.

Nitric acid

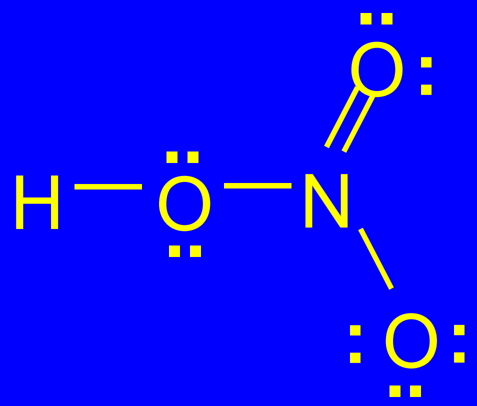
Formal charge of H



- We will calculate the formal charge for each atom in this Lewis structure.

Nitric acid

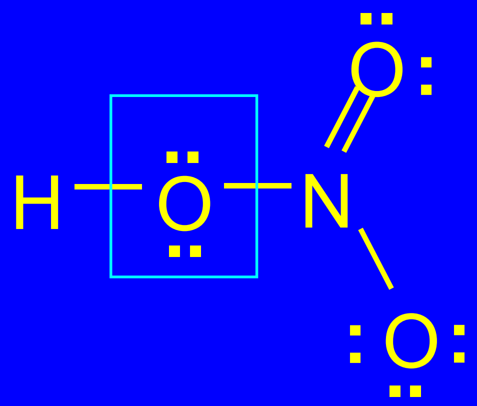
Formal charge of H



- Hydrogen shares 2 electrons with oxygen.
- Assign 1 electron to H and 1 to O.
- A neutral hydrogen atom has 1 electron.
- Therefore, the formal charge of H in nitric acid is 0.

Nitric acid

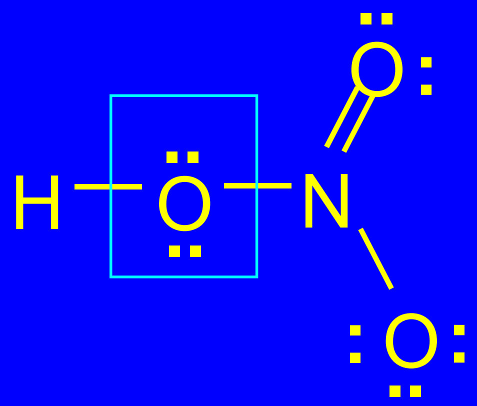
Formal charge of O



- Oxygen has 4 electrons in covalent bonds.
- Assign 2 of these 4 electrons to O.
- Oxygen has 2 unshared pairs. Assign all 4 of these electrons to O.
- Therefore, the total number of electrons assigned to O is 2 + 4 = 6.

Nitric acid

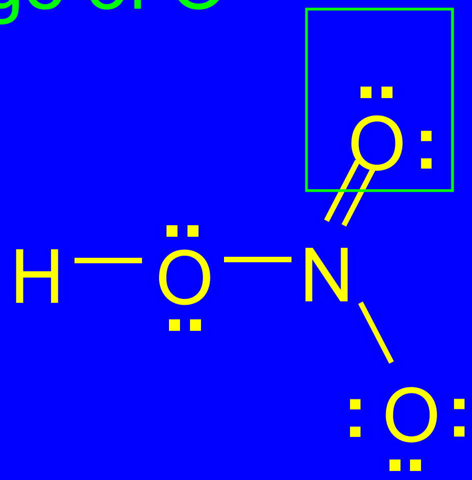
Formal charge of O



- Electron count of O is 6.
- A neutral oxygen has 6 electrons.
- Therefore, the formal charge of O is 0.

Nitric acid

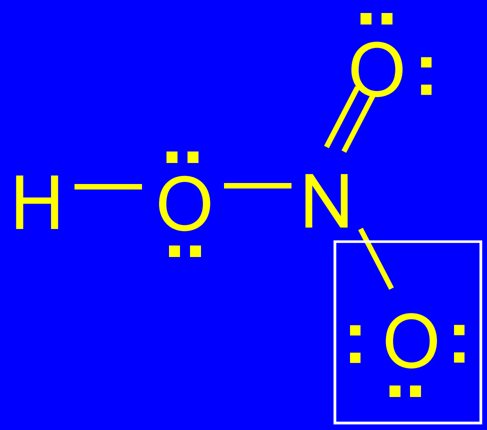
Formal charge of O



- Electron count of O is 6 (4 electrons from unshared pairs + half of 4 bonded electrons).
- A neutral oxygen has 6 electrons.
- Therefore, the formal charge of O is 0.

Nitric acid

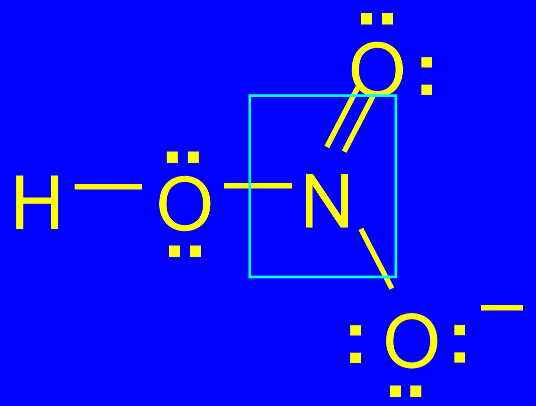
Formal charge of O



- Electron count of O is 7 (6 electrons from unshared pairs + half of 2 bonded electrons).
- A neutral oxygen has 6 electrons.
- Therefore, the formal charge of O is -1.

Nitric acid

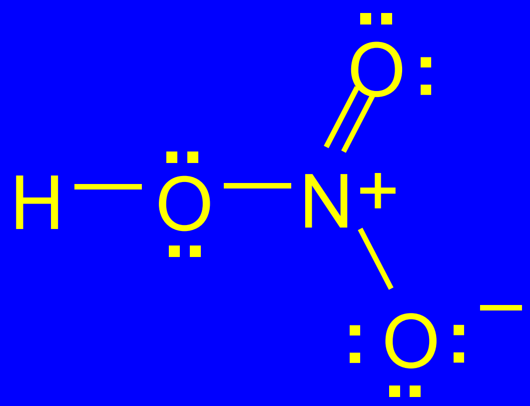
Formal charge of N



- Electron count of N is 4 (half of 8 electrons in covalent bonds).
- A neutral nitrogen has 5 electrons.
- Therefore, the formal charge of N is +1.

Nitric acid

Formal charges



- A Lewis structure is not complete unless formal charges (if any) are shown.

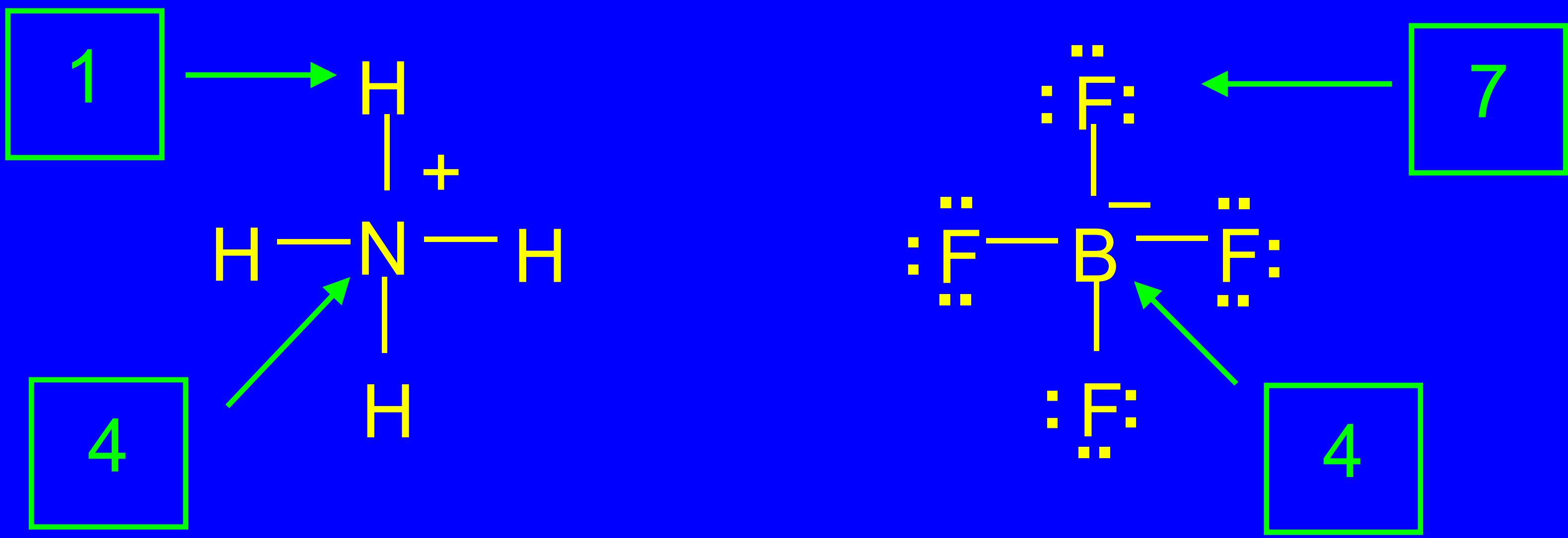
Formal Charge

An arithmetic formula for calculating formal charge.

Formal charge =

group number in periodic table	–	number of bonds	–	number of unshared electrons

"Electron counts" and formal charges in NH_4^+ and BF_4^-



3.5
Drawing Lewis Structures

Constitution

- The order in which the atoms of a molecule are connected is called its *constitution* or *connectivity*.
- The constitution of a molecule must be determined in order to write a Lewis structure.

Table 1.4 How to Write Lewis Structures

- Step 1:

  The molecular formula and the connectivity are determined by experiment.

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- Example:
  Methyl nitrite has the molecular formula CH_3NO_2. All hydrogens are bonded to carbon, and the order of atomic connections is CONO.

Table 1.4 How to Write Lewis Structures

- Step 2:

  Count the number of valence electrons. For a neutral molecule this is equal to the number of valence electrons of the constituent atoms.

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- Example (CH_3NO_2):

  Each hydrogen contributes 1 valence electron. Each carbon contributes 4, nitrogen 5, and each oxygen 6 for a total of 24.

Table 1.4 How to Write Lewis Structures

- Step 3:

  Connect the atoms by a covalent bond represented by a dash.

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  Connect the atoms by a covalent bond represented by a dash.

- Example:

  Methyl nitrite has the partial structure:

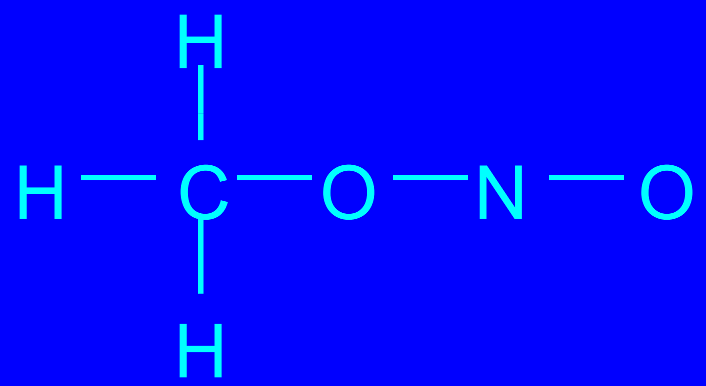

Table 1.4 How to Write Lewis Structures

- Step 4:

  Subtract the number of electrons in bonds from the total number of valence electrons.

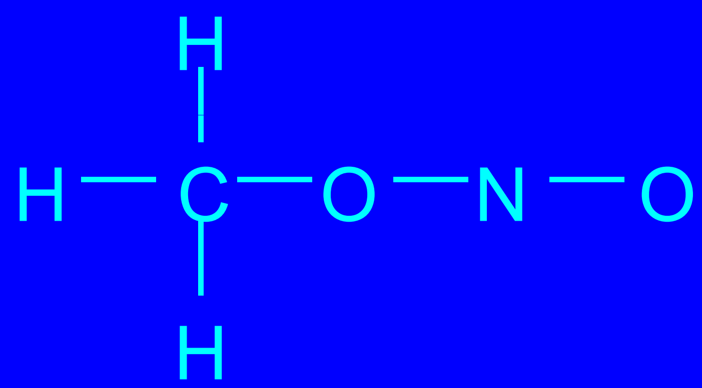

Table 1.4 How to Write Lewis Structures

- Step 4:
  Subtract the number of electrons in bonds from the total number of valence electrons.

- Example:
  24 valence electrons – 12 electrons in bonds.
  Therefore, 12 more electrons to assign.

Table 1.4 How to Write Lewis Structures

- Step 5:

  Add electrons in pairs so that as many atoms as possible have 8 electrons. Start with the most electronegative atom.

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- Example:

  The remaining 12 electrons in methyl nitrite are added as 6 pairs.

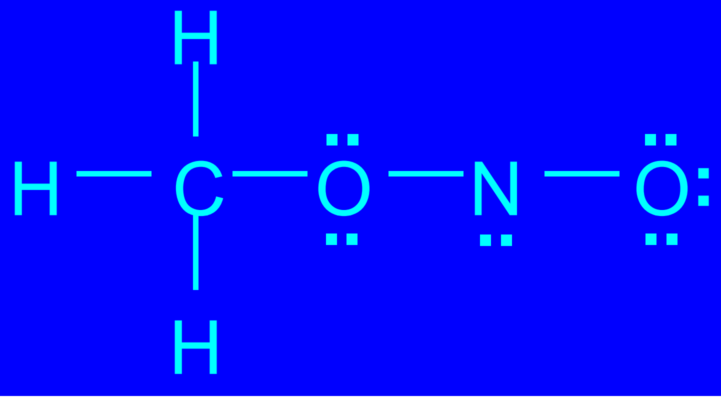


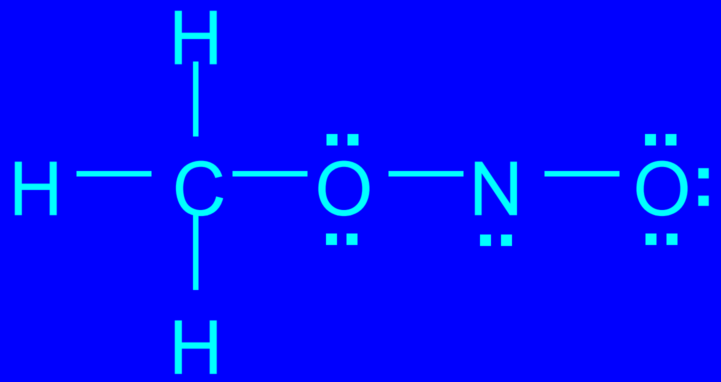
Table 1.4 How to Write Lewis Structures

- Step 6:

  If an atom lacks an octet, use electron pairs on an adjacent atom to form a double or triple bond.

- Example:

  Nitrogen has only 6 electrons in the structure shown.



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- Example:

  All the atoms have octets in this Lewis structure.

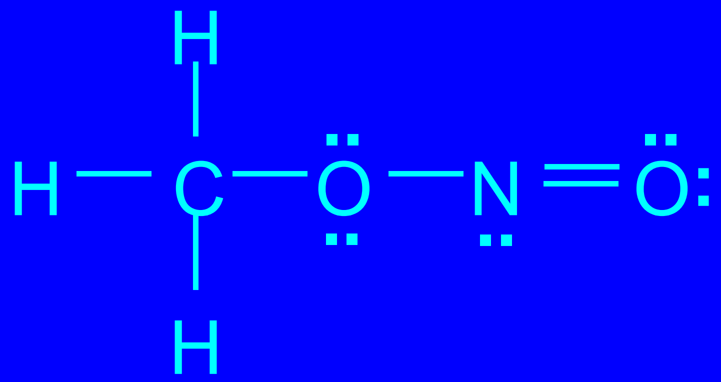

Table 1.4 How to Write Lewis Structures

- Step 7:

  Calculate formal charges.

- Example:

  None of the atoms possess a formal charge in this Lewis structure.

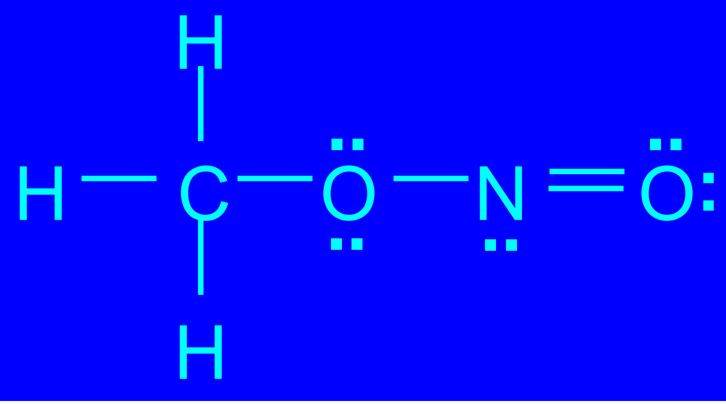

Table 1.4 How to Write Lewis Structures

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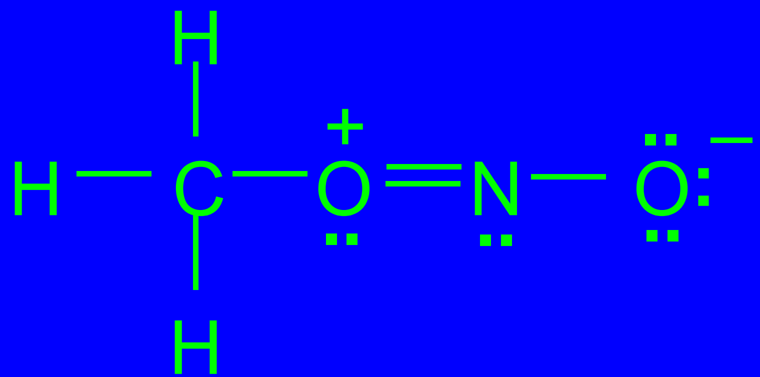
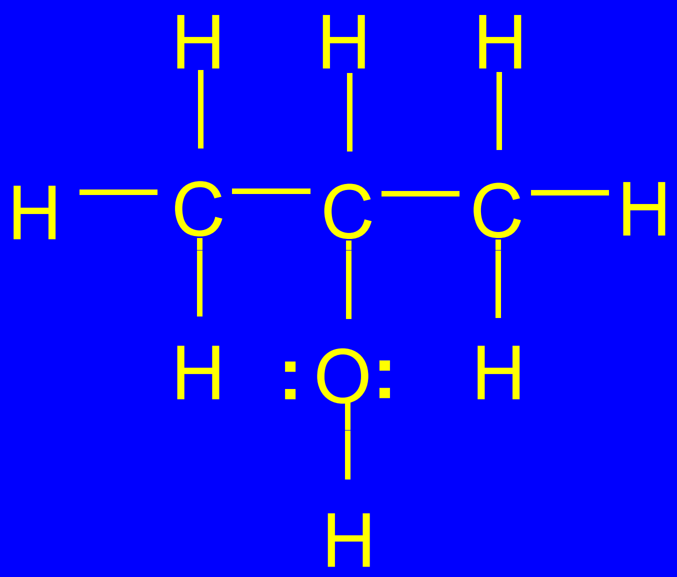
- Example:

  This structure has formal charges; is less stable Lewis structure.

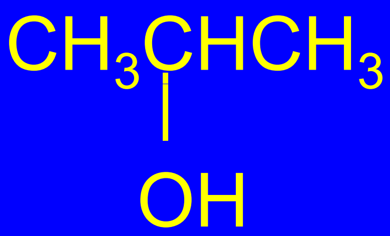


Condensed structural formulas

- Lewis structures in which many (or all) covalent bonds and electron pairs are omitted.



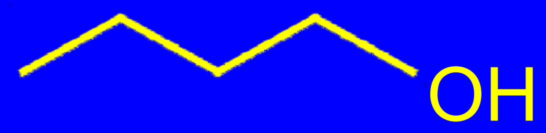
can be condensed to:



or $(CH_3)_2CHOH$

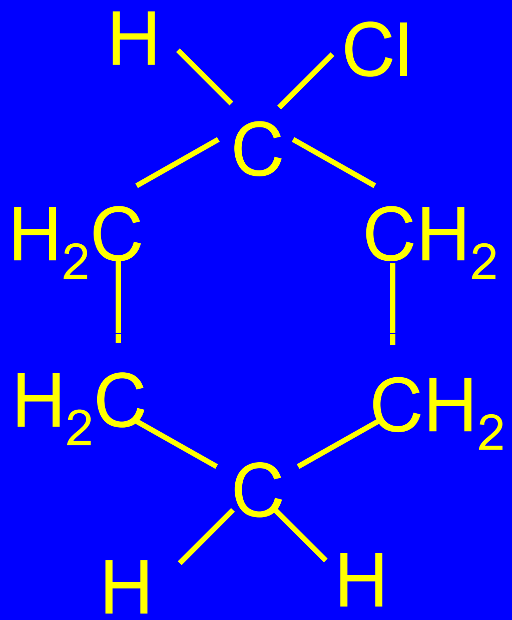
Bond-line formulas

$CH_3CH_2CH_2CH_3$ is shown as 

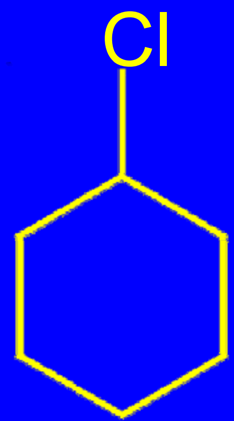
$CH_3CH_2CH_2CH_2OH$ is shown as 

- Omit atom symbols. Represent structure by showing bonds between carbons and atoms other than hydrogen.
- Atoms other than carbon and hydrogen are called *heteroatoms*.

Bond-line formulas



is shown as



- Omit atom symbols. Represent structure by showing bonds between carbons and atoms other than hydrogen.
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3.5
Constitutional Isomers

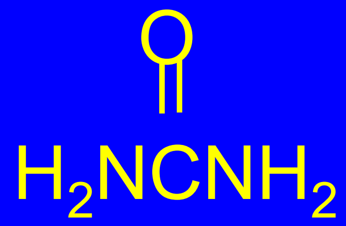
Constitutional isomers

- Isomers are different compounds that have the same molecular formula.
- Constitutional isomers are isomers that differ in the order in which the atoms are connected.
- An older term for constitutional isomers is “structural isomers.”

A Historical Note

NH_4OCN

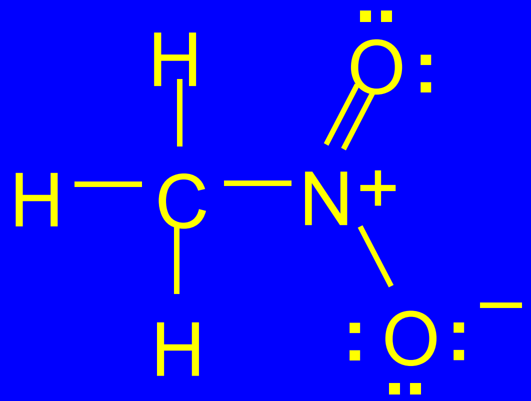
Ammonium cyanate



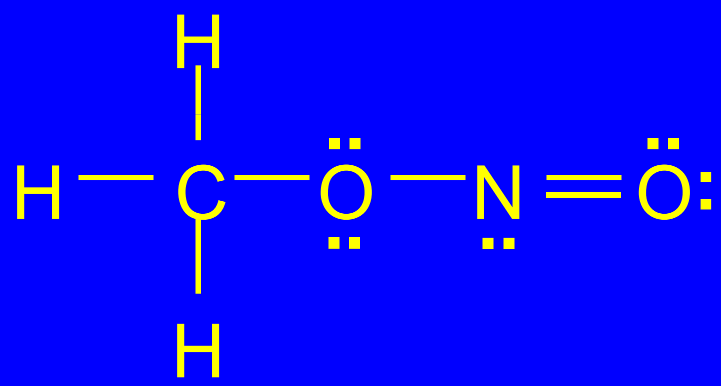
Urea

- In 1823 Friedrich Wöhler discovered that when ammonium cyanate was dissolved in hot water, it was converted to urea.
- Ammonium cyanate and urea are constitutional isomers of CH_4N_2O.
- Ammonium cyanate is "inorganic." Urea is "organic." Wöhler is credited with an important early contribution that helped overturn the theory of "vitalism."

Examples of constitutional isomers



Nitromethane



Methyl nitrite

- Both have the molecular formula CH_3NO_2 but the atoms are connected in a different order.

3.5

Resonance

Resonance

two or more acceptable octet Lewis structures

may be

written for certain compounds (or ions)

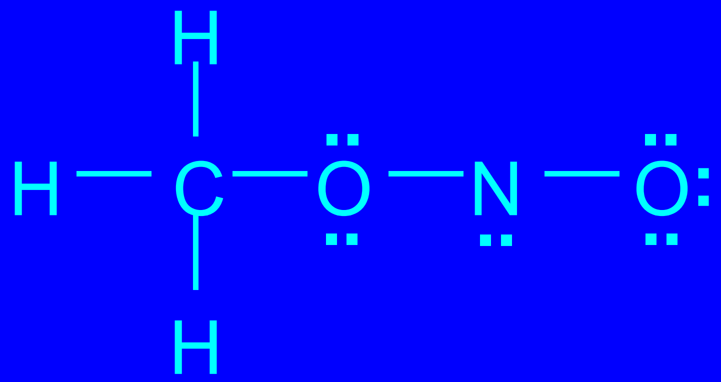
Table 1.4 How to Write Lewis Structures

- Step 6:

  If an atom lacks an octet, use electron pairs on an adjacent atom to form a double or triple bond.

- Example:

  Nitrogen has only 6 electrons in the structure shown.



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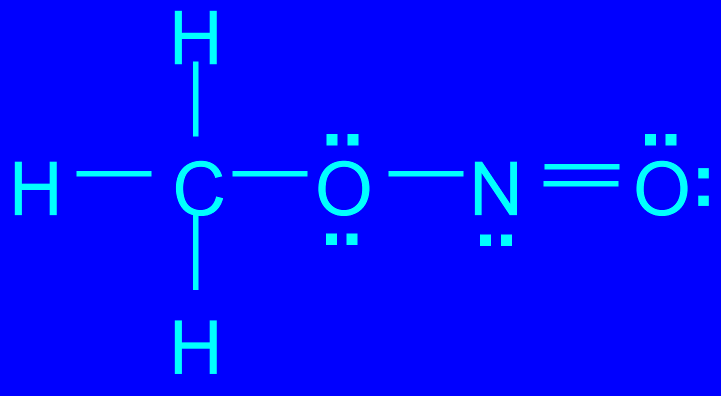

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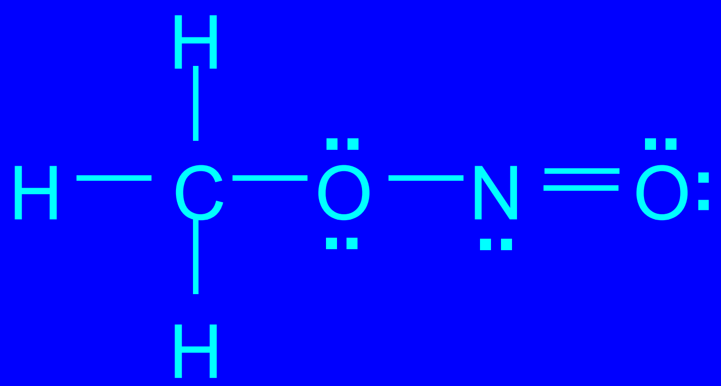

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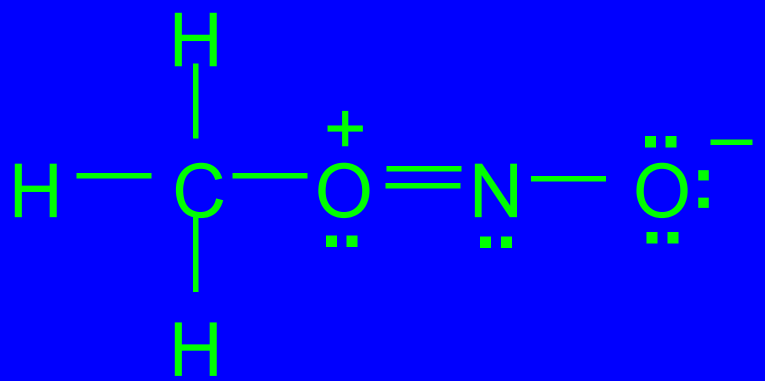
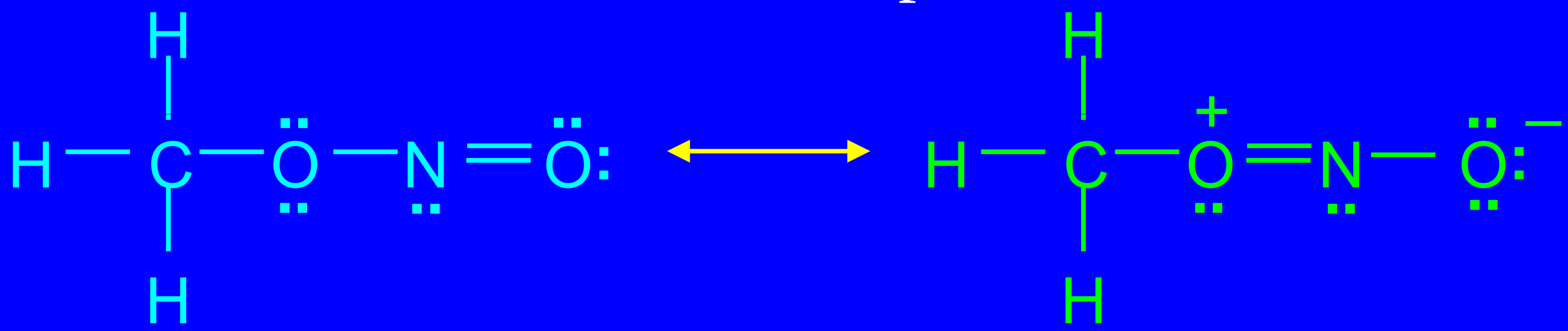


Resonance Structures of Methyl Nitrite

- same atomic positions
- differ in electron positions



more stable Lewis structure

less stable Lewis structure

Resonance Structures of Methyl Nitrite

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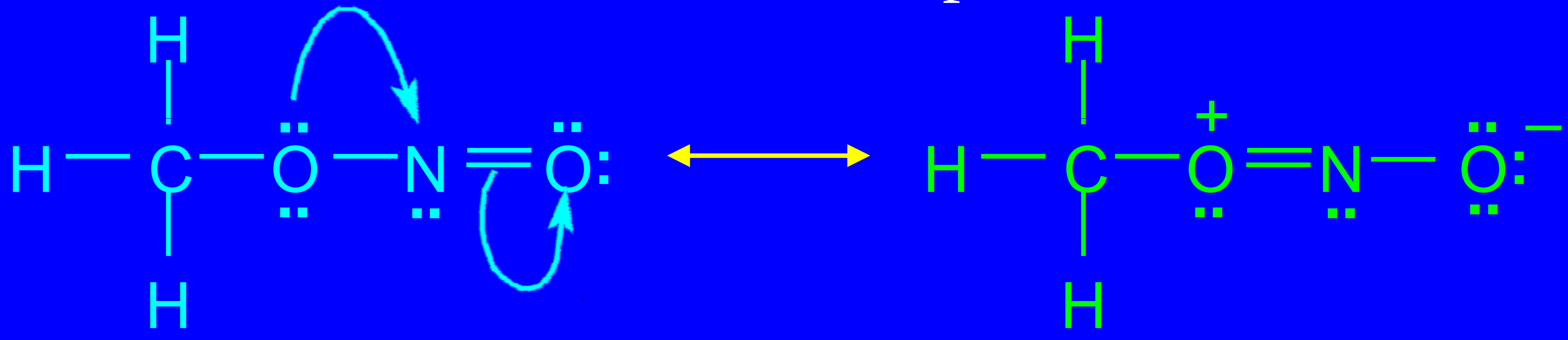
- differ in electron positions



more stable Lewis structure

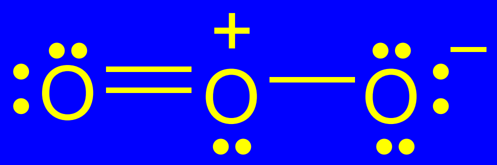
less stable Lewis structure

Why Write Resonance Structures?

- Electrons in molecules are often delocalized between two or more atoms.
- Electrons in a single Lewis structure are assigned to specific atoms-a single Lewis structure is insufficient to show electron delocalization.
- Composite of resonance forms more accurately depicts electron distribution.

Example

- Ozone (O_3)
  - Lewis structure of ozone shows one double bond and one single bond



Expect: one short bond and one long bond

Reality: bonds are of equal length (128 pm)

Example

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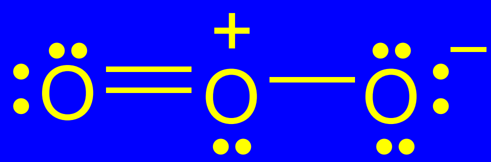
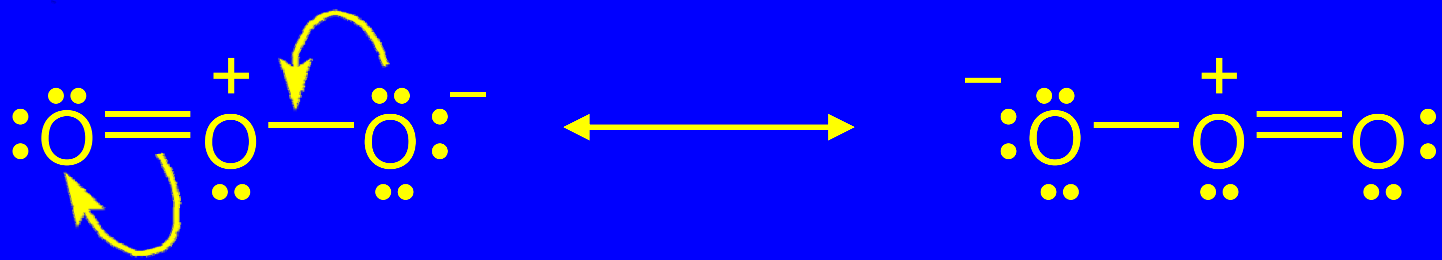
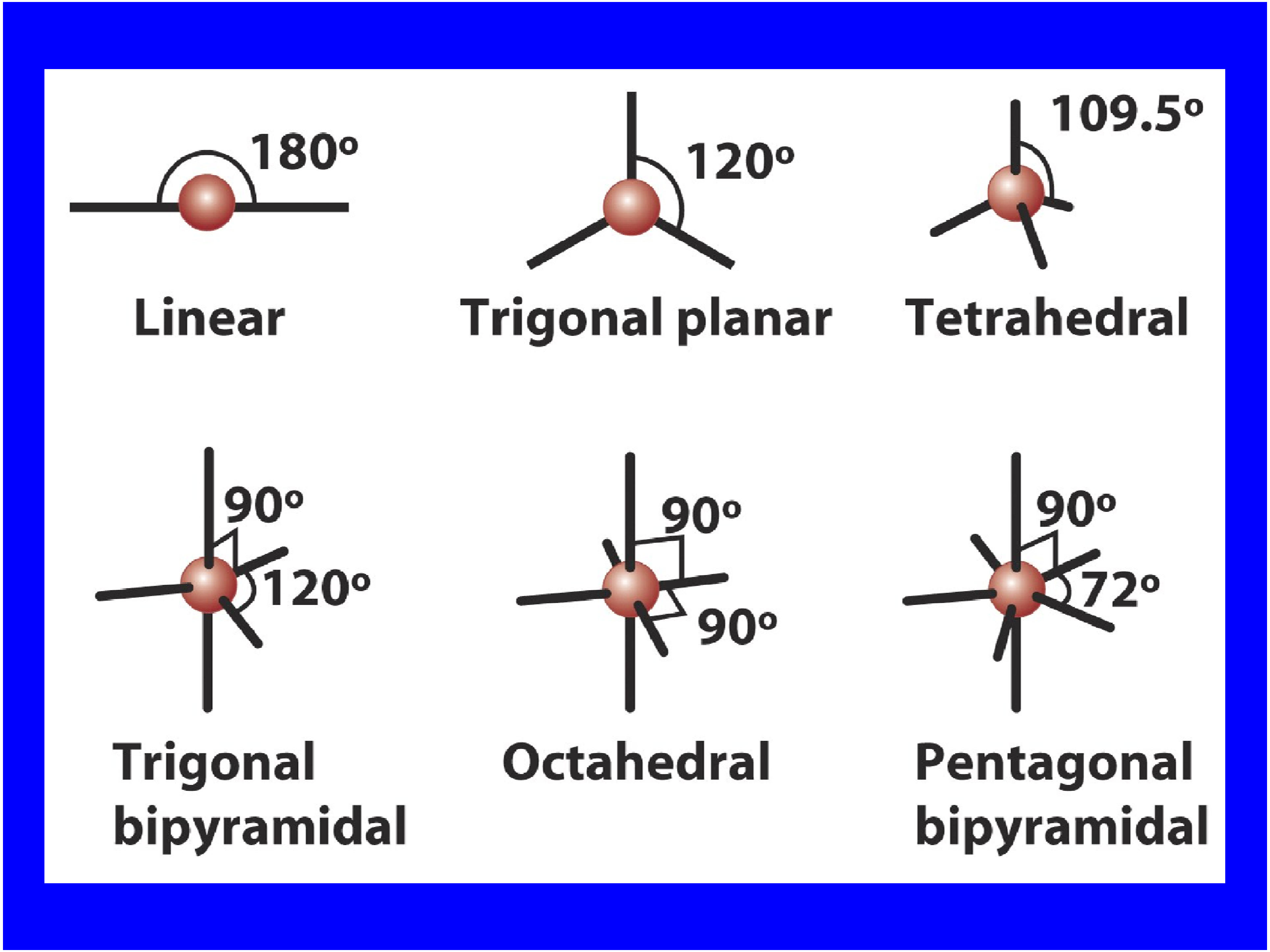
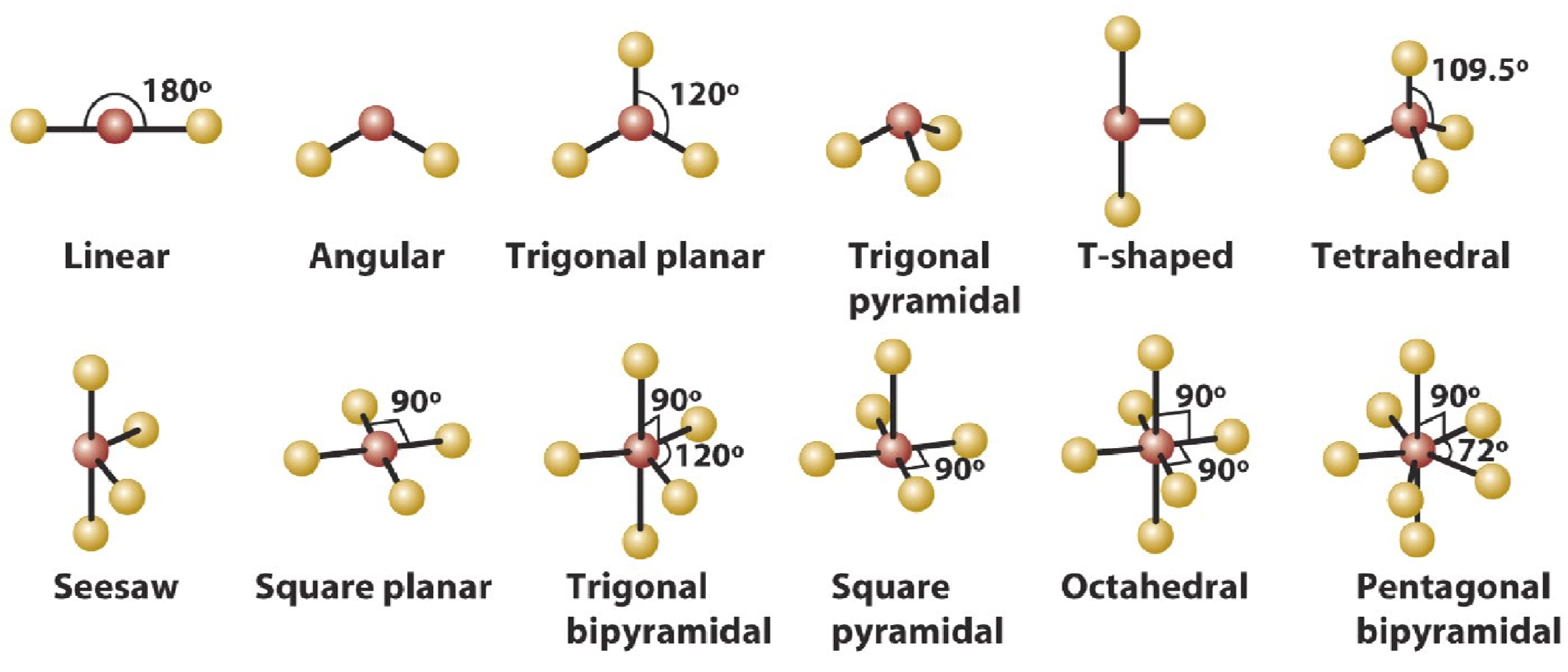


Resonance:



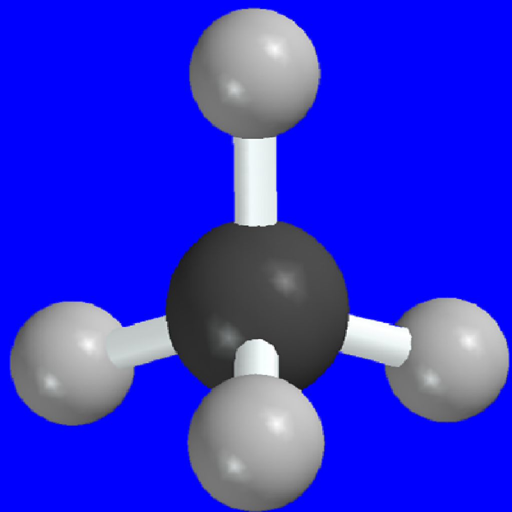
3.7
The Shapes of Some Simple Molecules


180°
Linear
120°
Trigonal planar
109.5°
Tetrahedral
90°
120°
Trigonal bipyramidal
90°
90°
Octahedral
90°
72°
Pentagonal bipyramidal


180°
Linear
Angular
120°
Trigonal planar
Trigonal pyramidal
T-shaped
109.5°
Tetrahedral
Seesaw
90°
Square planar
90°
120°
Trigonal bipyramidal
90°
Square pyramidal
90°
90°
Octahedral
90°
72°
Pentagonal bipyramidal

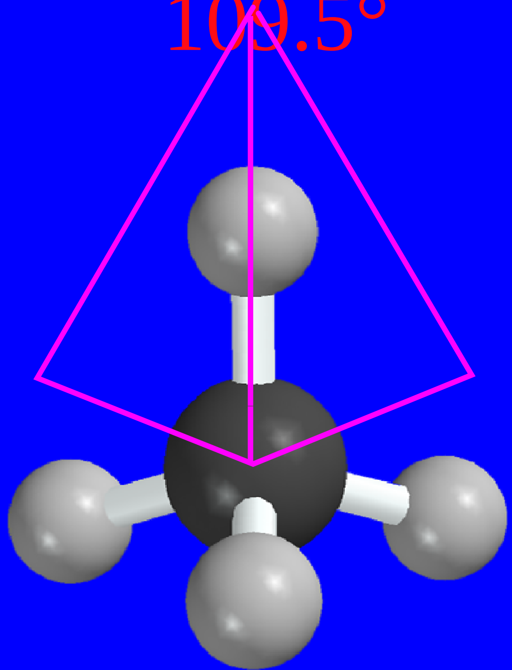
Methane

- tetrahedral geometry
- H—C—H angle = 109.5°



Methane

- tetrahedral geometry
- each H—C—H angle = 109.5°

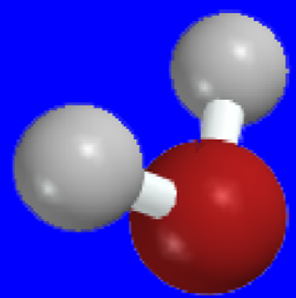


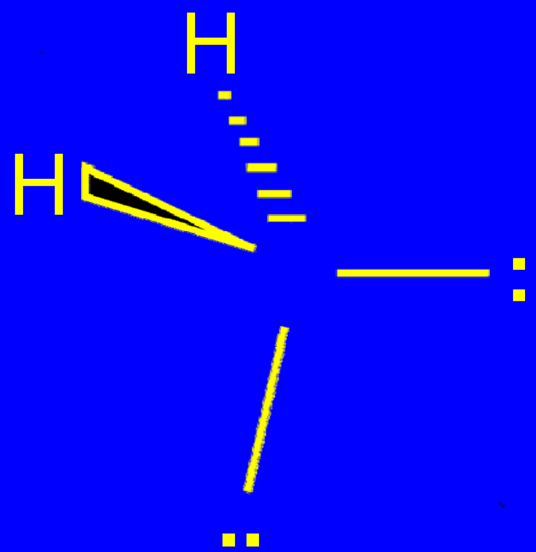
Valence Shell Electron Pair Repulsions

- The most stable arrangement of groups attached to a central atom is the one that has the maximum separation of electron pairs (bonded or nonbonded).

Water

- bent geometry
- H—O—H angle = 105°

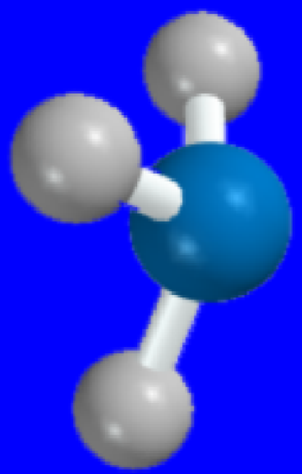


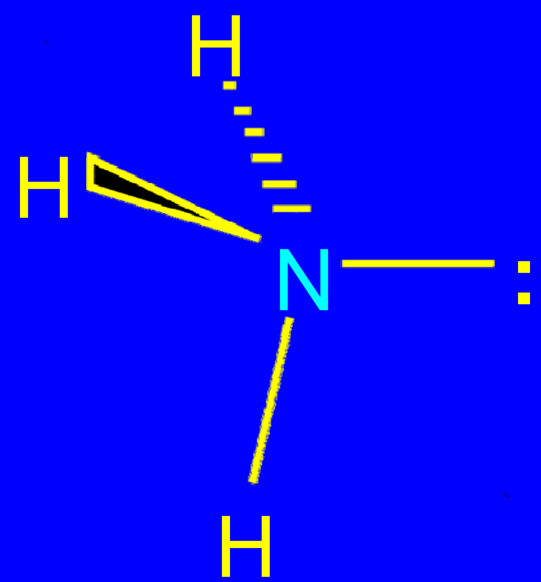


but notice the tetrahedral arrangement of electron pairs

Ammonia

- trigonal pyramidal geometry
- H—N—H angle = 107°

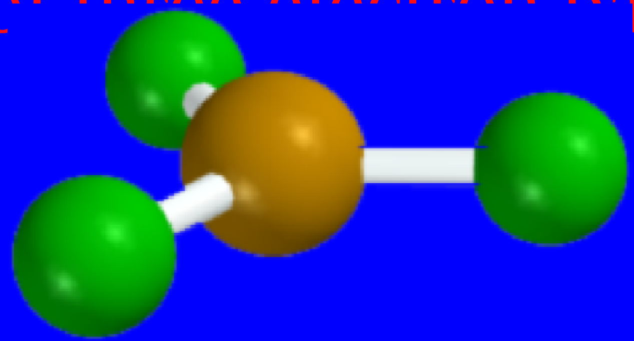




but notice the tetrahedral arrangement of electron pairs

Boron Trifluoride

- F—B—F angle = 120°
- trigonal planar geometry allows for maximum separation of three electron pairs

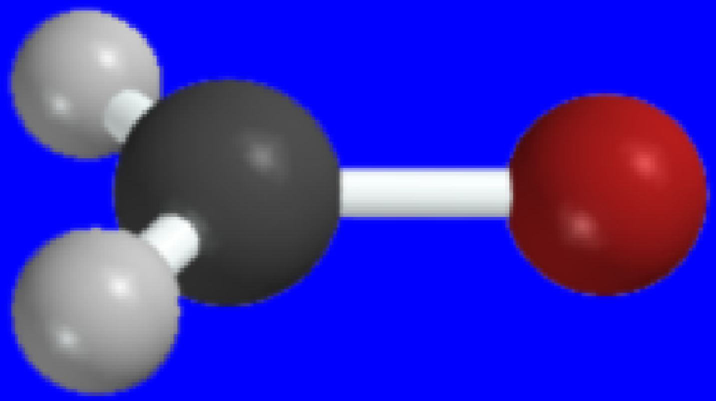


Multiple Bonds

- Four-electron double bonds and six-electron triple bonds are considered to be similar to a two-electron single bond in terms of their spatial requirements.

Formaldehyde: $CH_2{=}O$

- H—C—H and H—C—O angles are close to 120°
- trigonal planar geometry



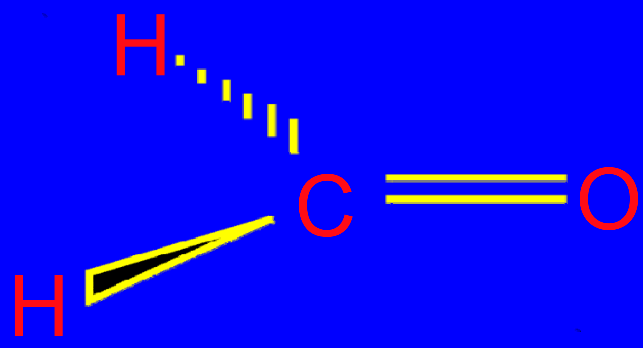
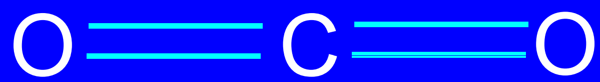
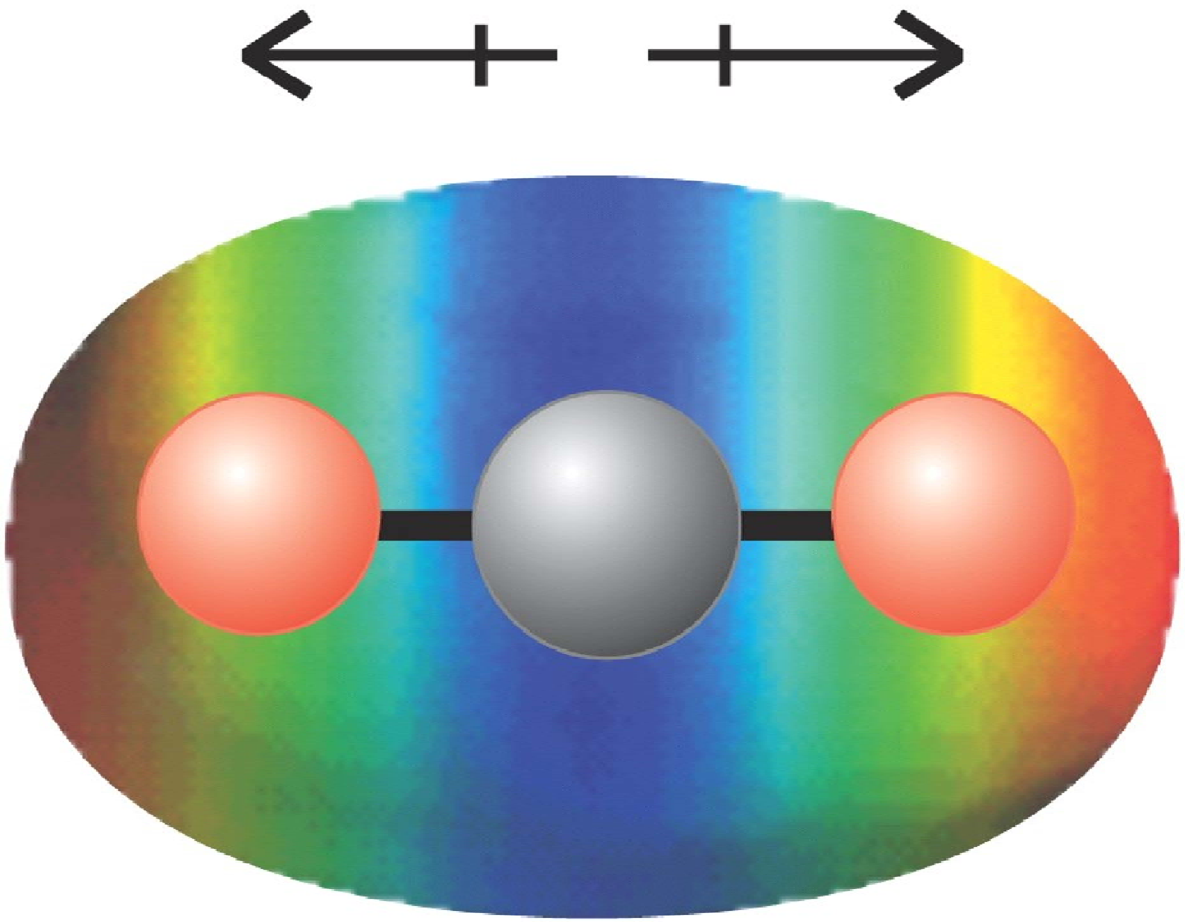


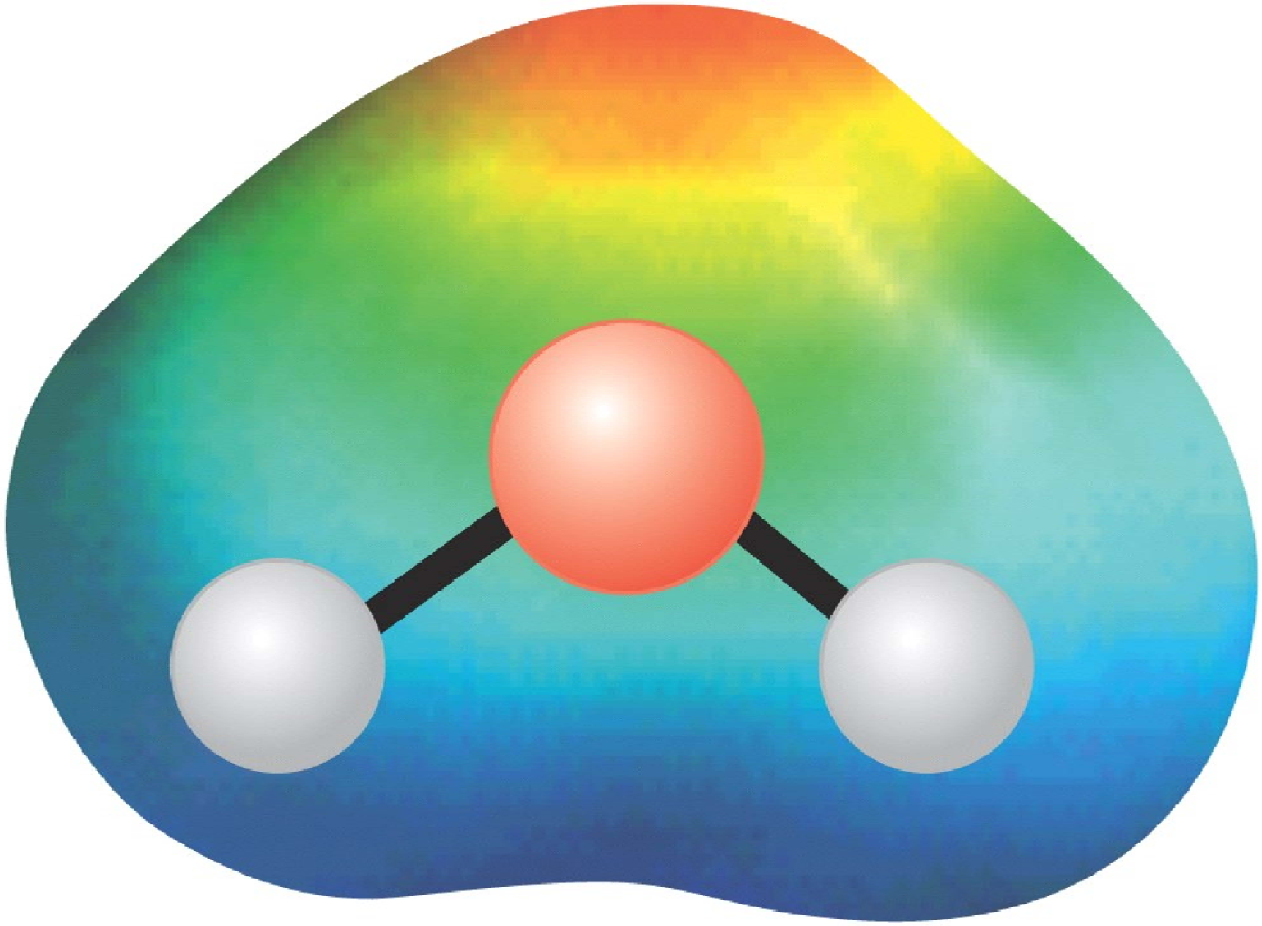
Figure 1.12: Carbon Dioxide

- O—C—O angle = 180°
- linear geometry





27 **Carbon dioxide, CO_2**



28 Water, H_2O

3.7: Polar Covalent Bonds and Electronegativity

Electronegativity

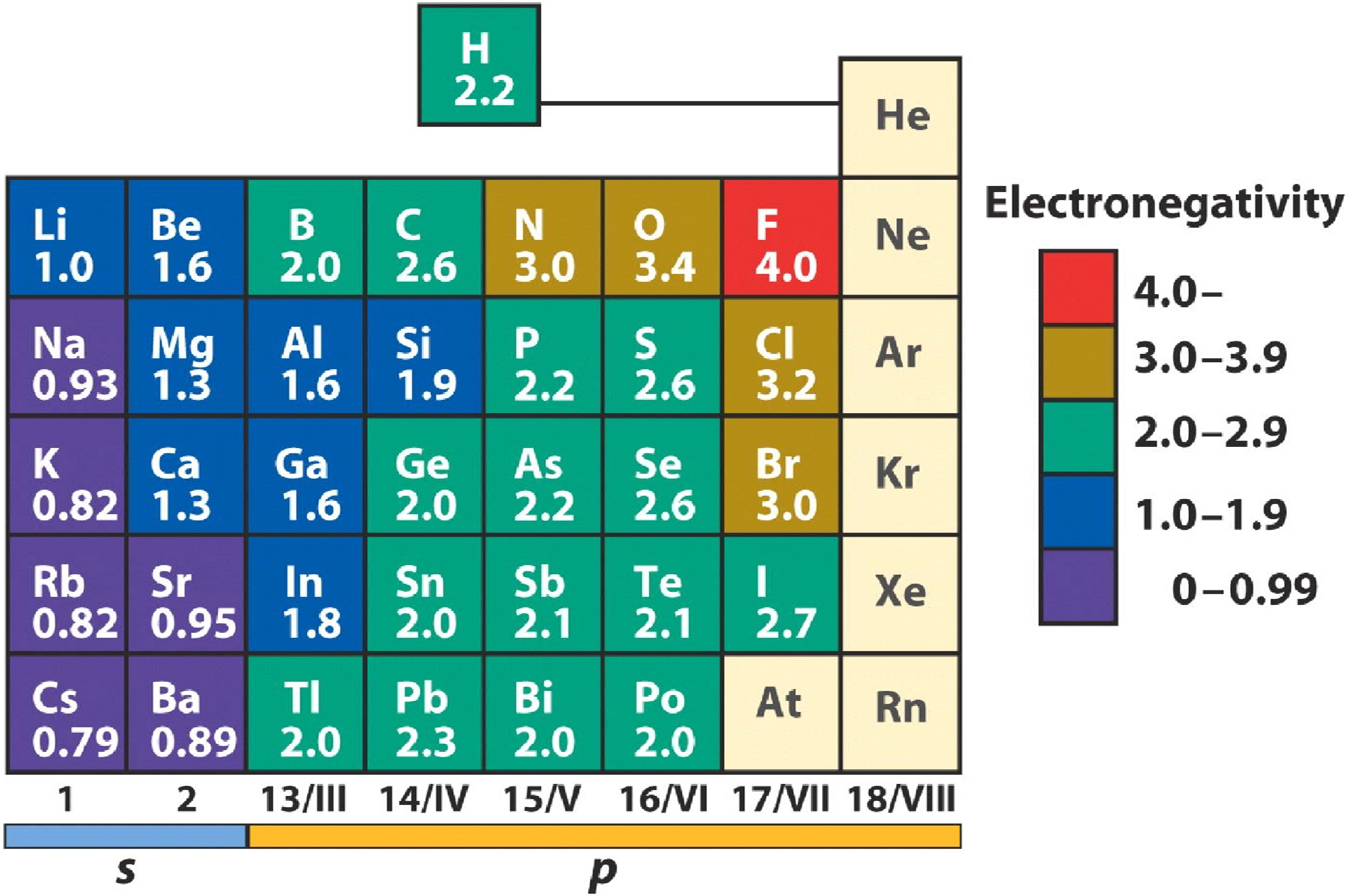
Electronegativity is a measure of an element to attract electrons toward itself when bonded to another element.

- An electronegative element attracts electrons.
- An electropositive element releases electrons.

Pauling Electronegativity Scale

Li	**Be**	**B**	**C**	**N**	**O**	**F**
1.0	**1.5**	**2.0**	**2.5**	**3.0**	**3.5**	**4.0**
Na	**Mg**	**Al**	**Si**	**P**	**S**	**Cl**
0.9	**1.2**	**1.5**	**1.8**	**2.1**	**2.5**	**3.0**

- Electronegativity increases from left to right in the periodic table.
- Electronegativity decreases going down a group.



H 2.2
He
Li 1.0
Be 1.6
B 2.0
C 2.6
N 3.0
O 3.4
F 4.0
Ne
Na 0.93
Mg 1.3
Al 1.6
Si 1.9
P 2.2
S 2.6
Cl 3.2
Ar
K 0.82
Ca 1.3
Ga 1.6
Ge 2.0
As 2.2
Se 2.6
Br 3.0
Kr
Rb 0.82
Sr 0.95
In 1.8
Sn 2.0
Sb 2.1
Te 2.1
I 2.7
Xe
Cs 0.79
Ba 0.89
Tl 2.0
Pb 2.3
Bi 2.0
Po 2.0
At
Rn
1
2
13/III
14/IV
15/V
16/VI
17/VII
18/VIII
s
p
Electronegativity
4.0–
3.0–3.9
2.0–2.9
1.0–1.9
0–0.99

Generalization

- The greater the difference in electronegativity between two bonded atoms; the more polar the bond.

H—H   :F—F: :N≡N:

nonpolar bonds connect atoms of the same electronegativity

Generalization

- The greater the difference in electronegativity
  between two bonded atoms; the more polar the
  bond.

$\overset{\delta+}{H}-\overset{\delta-}{\ddot{\underset{..}{F}}}:$ $\quad$ $\overset{\delta+}{H}-\overset{\delta-}{\ddot{\underset{..}{O}}}-\overset{\delta+}{H}$ $\quad$ $:\overset{\delta-}{\underset{..}{O}}=\overset{\delta+}{C}=\overset{\delta-}{\underset{..}{O}}:$

polar bonds connect atoms of

different electronegativity

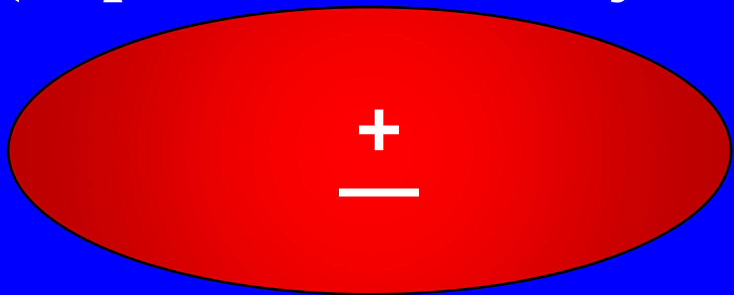
3.7
Molecular Dipole Moments

Dipole Moment

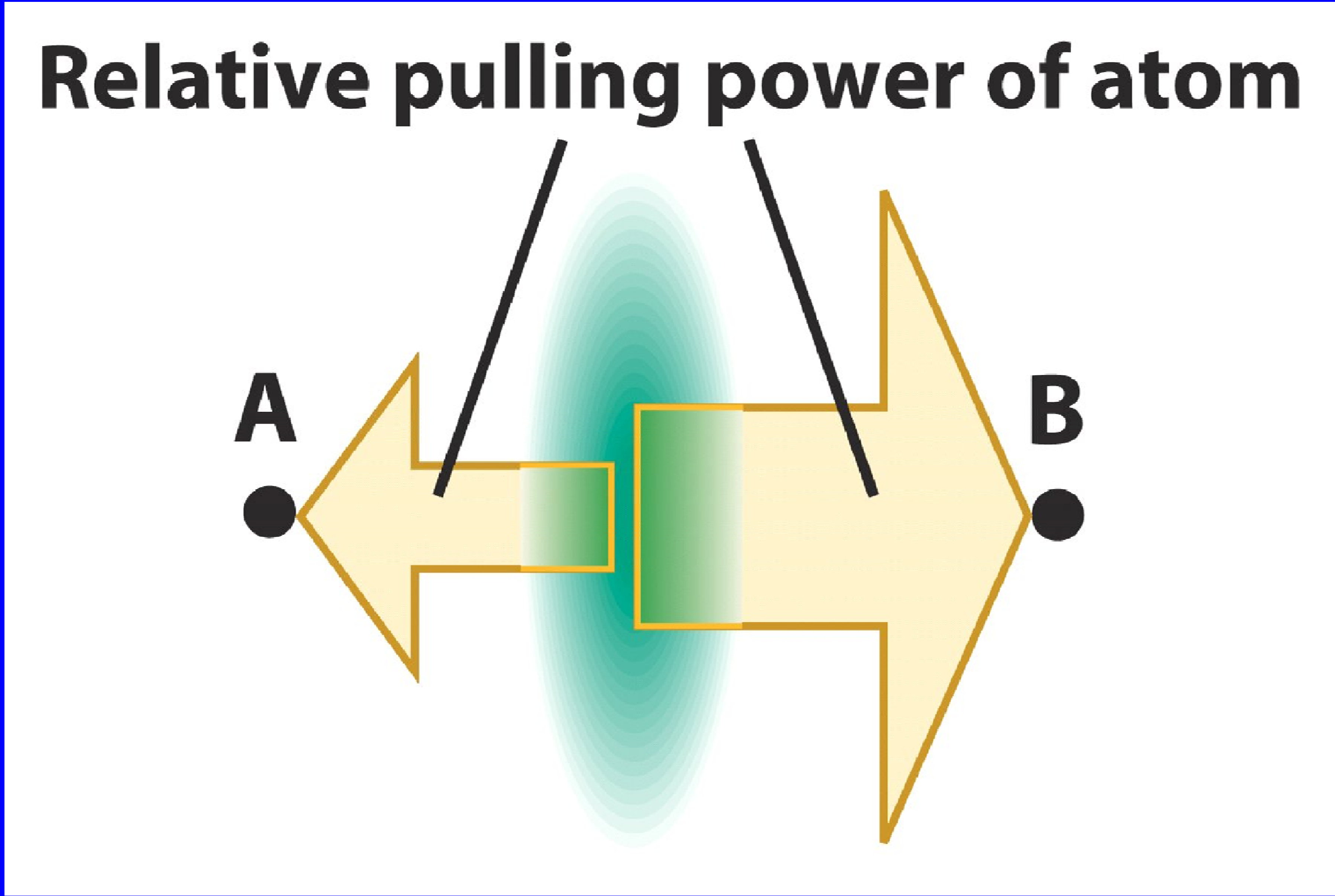
- A substance possesses a dipole moment if its centers of positive and negative charge do not coincide.

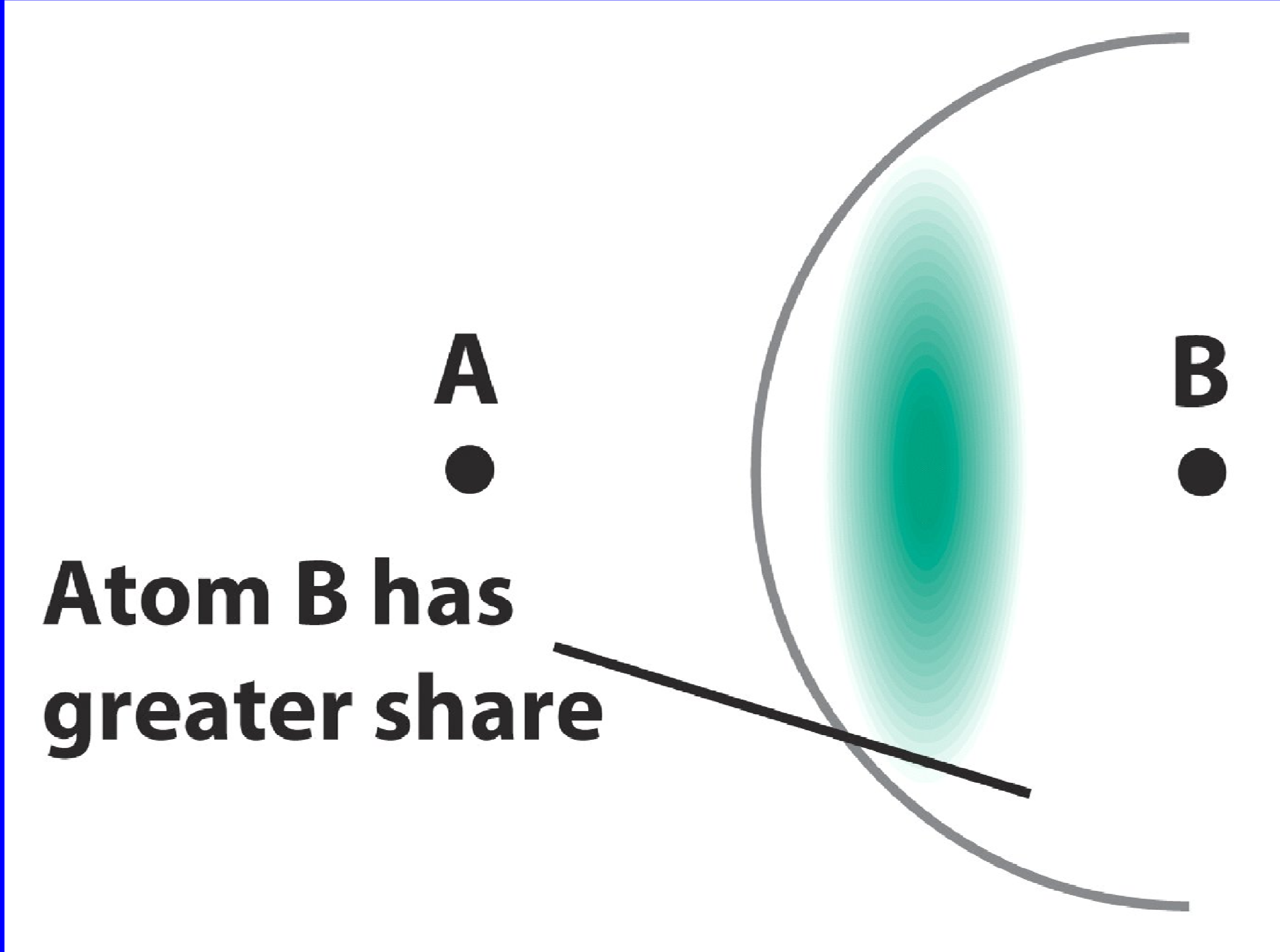
$$\mu = e \times d$$

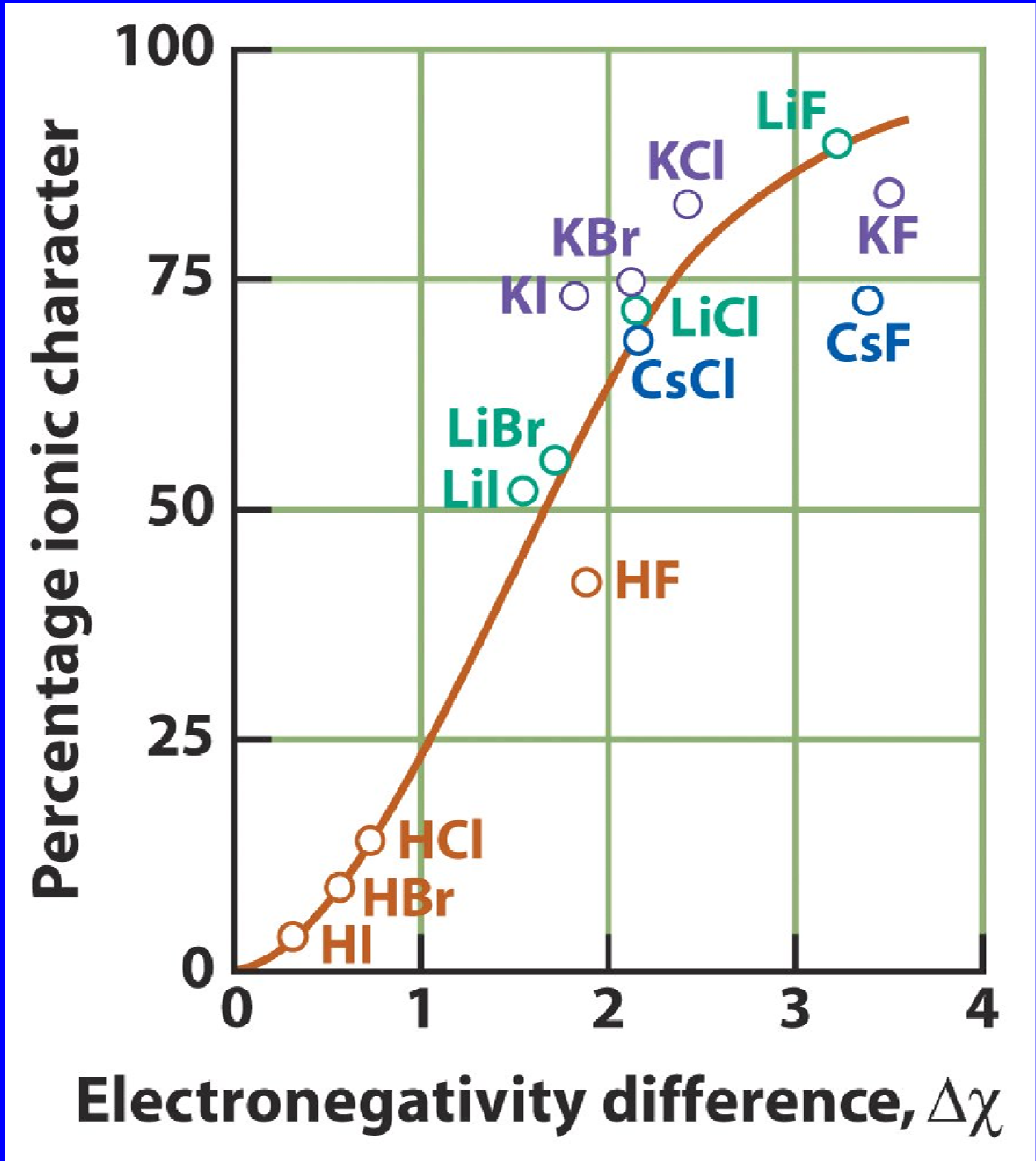
- (expressed in Debye units)



not polar


Relative pulling power of atom
A
B


A
B
Atom B has
greater share



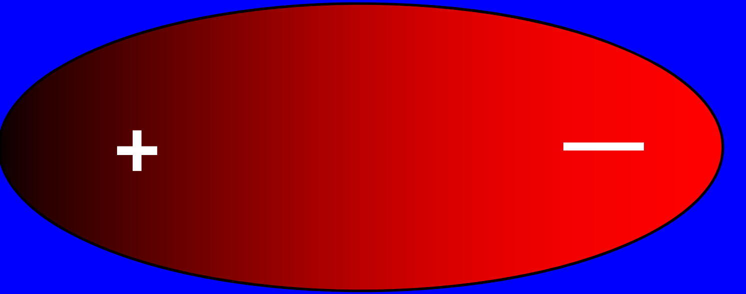
Percentage ionic character
100
75
50
25
0
0
1
2
3
4
Electronegativity difference, Δχ
LiF
KCl
KF
KBr
KI
LiCl
CsF
CsCl
LiBr
LiI
HF
HCl
HBr
HI

Dipole Moment

- A substance possesses a dipole moment if its centers of positive and negative charge do not coincide.

$$\mu = e \times d$$

- (expressed in Debye units)



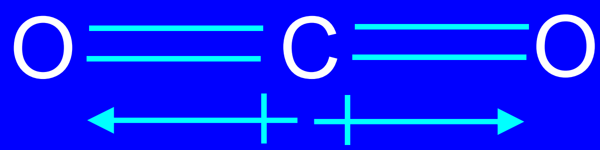
polar

Molecular Dipole Moments

$$\overset{\delta -}{O}=\overset{\delta +}{C}=\overset{\delta -}{O}$$

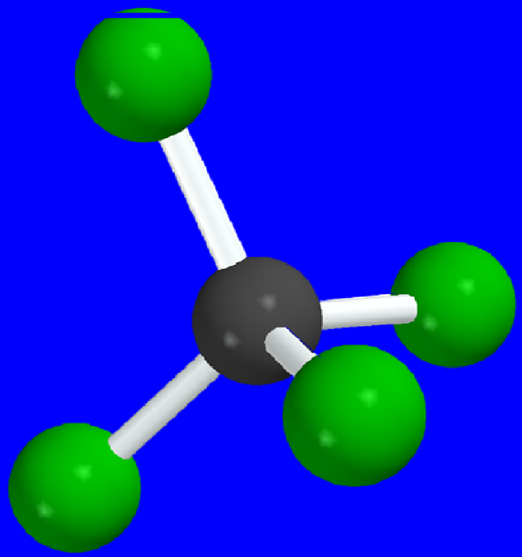
- molecule must have polar bonds
  - necessary, but not sufficient
- need to know molecular shape
  - because individual bond dipoles can cancel

Molecular Dipole Moments



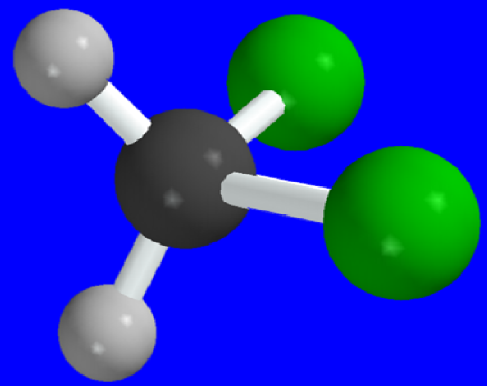
Carbon dioxide has no dipole moment; $\mu = 0$ D

Comparison of Dipole Moments



Carbon tetrachloride

$\mu = 0$ D

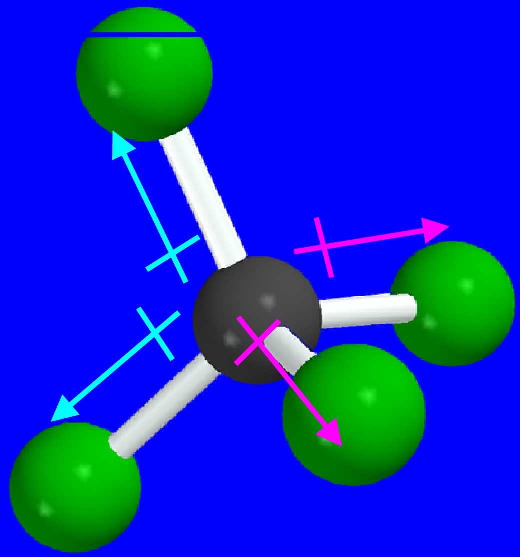


Dichloromethane

$\mu = 1.62$ D

Carbon tetrachloride

Resultant of these two bond dipoles is



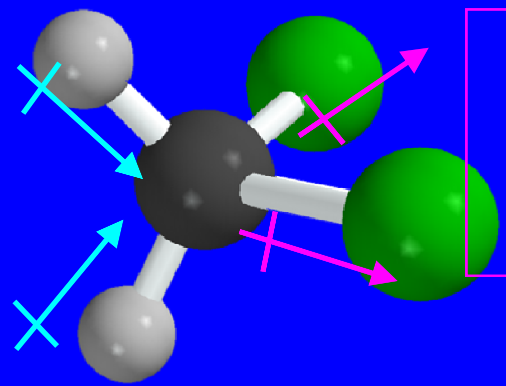
Resultant of these two bond dipoles is

μ = 0 D

Carbon tetrachloride has no dipole moment because all of the individual bond dipoles cancel.

Dichloromethane

Resultant of these two bond dipoles is



Resultant of these two bond dipoles is

μ = 1.62 D

The individual bond dipoles do not cancel in dichloromethane; it has a dipole moment.